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PROBLEM STATEMENT: Implementing a real-time undo/redo system for a text editing application using a Stack data structure. The system should support the following operations:

- Make a Change: A new change to the document is made.
- Undo Action: Revert the most recent change and store it for potential redo.
- Redo Action: Reapply the most recently undone action.
- Display Document State: Show the current state of the document after undoing or redoing an action

'''

```
class TextEditor:
```

```
def __init__(self):
```

```
    self.document=""
```

```
    self.Undo_stack=[]
```

```
    self.Redo_stack=[]
```

```
def make_change(self,change):
```

```
    self.Undo_stack.append(self.document)
```

```
    self.document+=change
```

```
    self.Redo_stack.clear()
```

```
    print("\n change made.")
```

```
    self.display_content()
```

```
def undo_action(self):
```

```
    if self.Undo_stack:
```

```
        self.Redo_stack.append(self.document)

        self.document=self.Undo_stack.pop()

        print("\n Undo performed")

    else:

        print("\n no more action to Undo")


def redo_action(self):

    if self.Redo_stack:

        self.Undo_stack.append(self.document)

        self.document=self.Redo_stack.pop()

        print("\n redo performed")

    else:

        print("\n no more action to Redo")


def display_content(self):

    print("the original document is :",self.document)


def menu(self):

    while True:

        print("===menu===")

        print("1.Make a change")

        print("2.Undo")

        print("3.Redo")

        print("4.Display")

        print("5.Exit")

        choice=int(input("Enter your choice:"))
```

```
if choice==1:
    change=input("Text to be added:")
    self.make_change(change)
elif choice==2:
    self.undo_action()
elif choice==3:
    self.redo_action()
elif choice==4:
    self.display_content()
elif choice==5:
    print(exit)
```

```
Editor=TextEditor()
```

```
Editor.menu()
```

```
'''OUTPUT
```

```
gescoe@gescoe-OptiPlex-3020:~/SE divB $ python3 Stack.py
```

```
===menu===
```

```
1.Make a change
```

```
2.Undo
```

```
3.Redo
```

```
4.Display
```

```
5.Exit
```

```
Enter your choice:1
```

```
Text to be added:aaa
```

change made.

the original document is : aaa

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:1

Text to be added:bbb

change made.

the original document is : aaabbb

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:2

Undo performed

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:2

Undo performed

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:3

redo performed

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:4

the original document is : aaa

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:4

the original document is : aaa

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:5

Use exit() or Ctrl-D (i.e. EOF) to exit

===menu===

1.Make a change

2.Undo

3.Redo

4.Display

5.Exit

Enter your choice:

'''